

Class Project Phase I: CSE 460

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Actors

(SunDevil Connect – College Event & Club Management System)

Student User

- **Description:**
A student who uses the SunDevil Connect system to discover campus events and student clubs. The student can browse and filter events, view event and club details, join clubs, and register for events. The student must log in to access personalized features such as event registration and club membership.

Club Leader

- **Description:**
A club leader is a student with additional privileges who manages a student organization. The club leader can create and manage events, approve or reject club membership requests, and post announcements for club members. Club leaders must log in to access the club dashboard.

System Administrator (College Admin)

- **Description:**
A system administrator oversees the entire SunDevil Connect platform. The administrator approves new club registrations, manages flagged or inappropriate content, and monitors overall system activity to ensure proper operation of the platform.

Authentication System

- **Description:**
An external authentication system (such as ASU Single Sign-On) that verifies user credentials and ensures secure access to the SunDevil Connect system.

Use Cases

Login

Description:

Allows users (students, club leaders, and administrators) to log into the system. The login use case includes displaying the login screen and verifying user credentials through the authentication system.

- **Display Login Screen:**
The system displays the login screen for the user to enter credentials.
- **Authenticate User:**
The system verifies the user's credentials using the authentication system.

Browse Events

Description:

Allows students to browse upcoming campus events organized by category such as music, technology, sports, social, and career-related events.

- **Display Event List:**
The system displays a list of upcoming events grouped by category.

Search & Filter Events

Description:

Allows students to refine the list of events by applying search filters to find events that meet specific criteria.

- **Apply Filters:**
The system filters events based on date, location, free/paid status, and popularity.

View Event Details

Description:

Allows students to view detailed information about a selected event, including the event description, date, time, location, and hosting club.

- **Display Event Information:**
The system displays the selected event's details.

Register for Event

Description:

Allows a student to register for an event after viewing its details. The system records the student's registration for the selected event.

- Record Event Registration:
The system saves the student's registration for the event.

Browse Clubs

Description:

Allows students to browse a list of student clubs available on campus along with basic club descriptions.

- Display Club List:
The system displays a list of available student clubs.

View Club Page

Description:

Allows students to view detailed information about a specific club, including the club description, members, announcements, and upcoming events.

- Display Club Information:
The system displays the selected club's details, events, and announcements.

Join Club

Description:

Allows a student to request membership in a selected club. The request is submitted to the club leader for review.

- Submit Membership Request:
The system sends the membership request to the club leader for approval.

Manage Club Dashboard

Description:

Allows club leaders to manage club-related activities through a centralized dashboard.

- Display Club Management Options:
The system displays tools for managing events, members, and announcements.

Create Event

Description:

Allows a club leader to create a new event for their club by entering event details into the system.

- Enter Event Details:
The club leader provides the event title, date, time, location, and category.
- Save Event:
The system saves the newly created event and makes it available to students.

Edit / Cancel Event

Description:

Allows a club leader to update event information or cancel an existing event.

- Modify Event Details:
The system updates the event with revised information.
- Cancel Event:
The system removes or marks the event as canceled.

Approve Club Members

Description:

Allows a club leader to review and approve pending club membership requests submitted by students.

- Review Membership Requests:
The system displays pending membership requests.
- Approve or Reject Request:
The club leader approves or rejects the request.

Post Announcements

Description:

Allows a club leader to post announcements or updates that are visible to club members.

- Create Announcement:
The club leader enters announcement content.
- Publish Announcement:
The system publishes the announcement to the club page.

Approve Club Registration

Description:

Allows the system administrator to review and approve new club registration requests.

- Review Club Submission:
The system displays new club registration details.
- Approve Club:
The system activates the club on the platform.

Manage Flagged Content

Description:

Allows the system administrator to review and manage content that has been flagged as inappropriate or reported by users.

- Review Flagged Content:
The system displays reported content.
- Remove or Resolve Content:
The administrator takes appropriate action.

Oversee System Activity

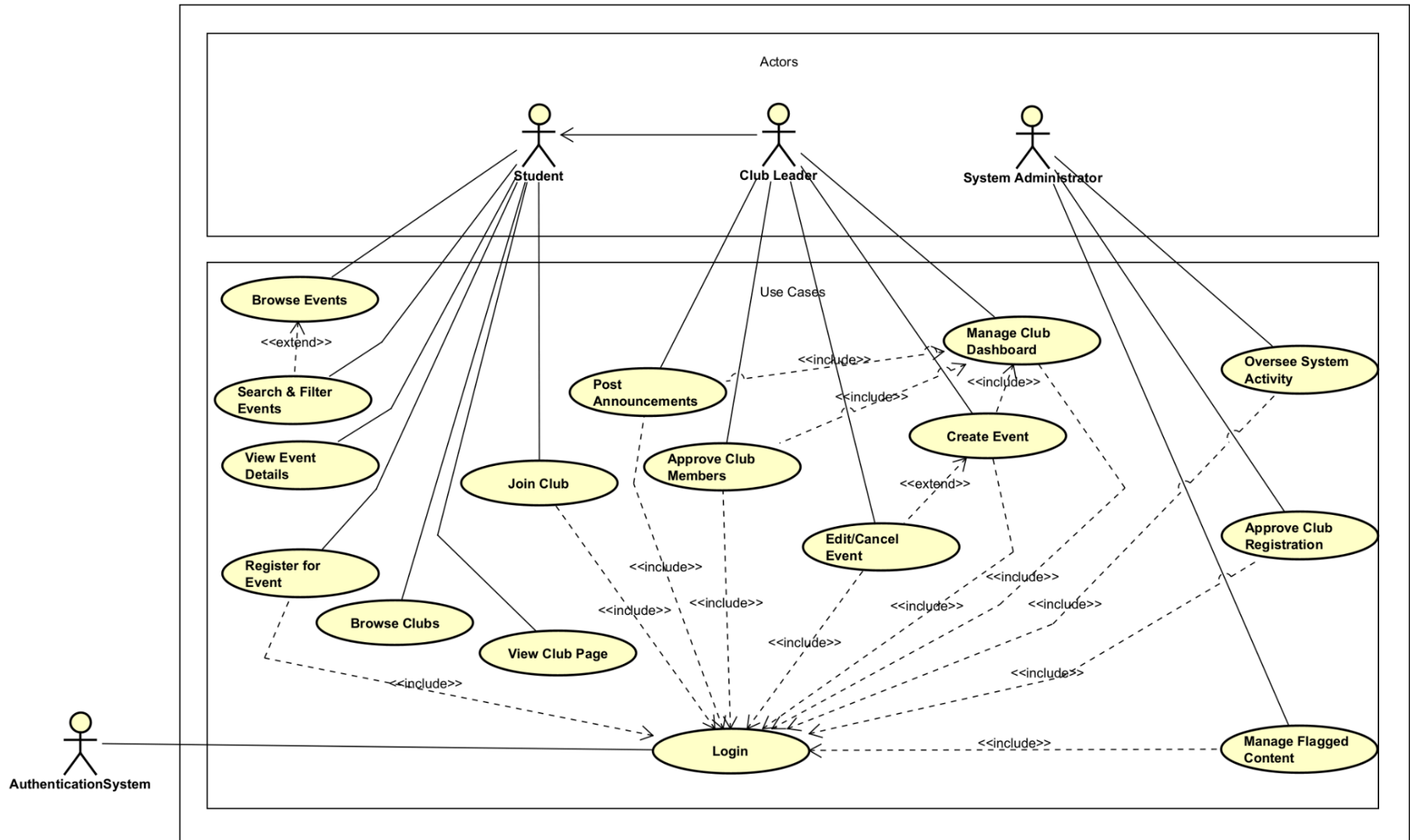
Description:

Allows the system administrator to monitor overall platform activity and ensure proper system operation.

- Monitor Platform Activity:
The system displays overall usage and administrative information.

Use Case Diagram

SunDevil Connect



Authentication (implied but required)

- ✓ Covered by:
- Login

Functional Requirement 1: Event Discovery
Students browse events by category
Students apply filters

- ✓ Covered by:
- Browse Events
- Search & Filter Events
- View Event Details

Functional Requirement 2: Club Membership

Users join clubs

Each club has a page with info, events, members, announcements

- ✓ Covered by:
- Browse Clubs
- View Club Page
- Join Club

Functional Requirement 3: Event Registration
Students can register for events

- ✓ Covered by:
- Register for Event

Functional Requirement 4: Club Dashboard (Leaders)

Create/manage events

Approve members

Post updates

- ✓ Covered by:
- Manage Club Dashboard
- Create Event
- Edit / Cancel Event
- Approve Club Members
- Post Announcements

Functional Requirement 5: Admin Panel

Oversee clubs

Approve new clubs

Manage flagged content

- ✓ Covered by:
- Approve Club Registration
- Manage Flagged Content
- Oversee System Activity

List of objects and their classification

Entity Objects

1. *User*
 - a. Admin
 - b. Student
2. Club
3. *Content*
 - a. Event
 - b. Announcement
4. Club Membership
5. Event Registration
6. Membership Request

Boundary Objects

1. Login Page
2. Event List View
3. Event Details View
4. Event Filter Panel
5. Club Page View
6. Club Browse View
7. Club Dashboard
8. Announcement List View
9. Admin Panel

Control Objects

1. Authentication Controller
2. Event Management Controller
3. Event Browsing Controller
4. Event Registration Controller
5. Club Browsing Controller
6. Club Membership Controller
7. Announcement Controller
8. Club Approval Controller
9. Content Moderation Controller

CRC Diagram

Entity Objects

Class: User

Description:

Represents a registered system user with authentication credentials and profile information.

Responsibilities / Collaborators:

- Store common user information → Authentication Controller
 - Provide base functionality for all users → Student, Admin
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Class: Student (extends User)

Description:

Represents a student user who interacts with clubs and events on the platform.

Responsibilities / Collaborators:

- Browse and filter events → Event Browsing Controller
 - Register for events → Event Registration Controller
 - Join clubs → Club Membership Controller
 - View club and event details → Club Browsing Controller, Event
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Class: Admin (extends User)

Description:

Represents an administrative user responsible for system oversight and moderation.

Responsibilities / Collaborators:

- Approve club registrations → Club Approval Controller
 - Manage flagged content → Content Moderation Controller
 - Monitor system activity → Content, Club
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Class: Club

Description:

Represents a student organization hosted on the platform.

Responsibilities / Collaborators:

- Store club information and metadata → Club Browsing Controller
 - Maintain list of members → Club Membership
 - Manage hosted events → Event, Event Management Controller
 - Publish announcements → Announcement, Announcement Controller
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Class: Event

Description:

Represents a scheduled campus event hosted by a club.

Responsibilities / Collaborators:

- Store event details → Event Management Controller
- Track registrations and capacity → Event Registration
- Provide event information → Event Browsing Controller
- Associate event with club → Club

Class: Announcement**Description:**

Represents an announcement published by a club.

Responsibilities / Collaborators:

- Store announcement content → Announcement Controller
- Publish announcements to members → Club

Class: Content**Description:**

Represents system content that may be reported or moderated.

Responsibilities / Collaborators:

- Track moderation status → Content Moderation Controller
- Support administrative review → Admin

Class: Club Membership**Description:**

Represents an approved membership between a student and a club.

Responsibilities / Collaborators:

- Link student to club → Student, Club
 - Track membership status → Club Membership Controller
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Class: Event Registration

Description:

Represents a student's registration for a specific event.

Responsibilities / Collaborators:

- Link student to event → Student, Event
 - Track registration status → Event Registration Controller
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Class: Membership Request

Description:

Represents a pending request from a student to join a club.

Responsibilities / Collaborators:

- Track approval or rejection → Club Membership Controller
- Associate request with student and club → Student, Club

Boundary Objects

Class: Login Page

Description:

Represents the user interface for authentication.

Responsibilities / Collaborators:

- Display login interface → User
 - Collect user credentials → Authentication Controller
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Class: Event List View

Description:

Represents the interface for browsing events.

Responsibilities / Collaborators:

- Display list of events → Event
 - Show categorized or filtered results → Event Browsing Controller
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Class: Event Details View**Description:**

Represents the interface for viewing detailed event information.

Responsibilities / Collaborators:

- Display event details → Event
 - Enable event registration → Event Registration Controller
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Class: Event Filter Panel**Description:**

Represents the interface for filtering events.

Responsibilities / Collaborators:

- Capture filter criteria → Event Browsing Controller
- Submit filters for processing → Event Browsing Controller

Class: Club Browse View

Description:

Represents the interface for browsing clubs.

Responsibilities / Collaborators:

- Display list of clubs → Club
 - Allow club selection → Club Browsing Controller
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Class: Club Page View**Description:**

Represents the interface for viewing a specific club.

Responsibilities / Collaborators:

- Display club details → Club
 - Display events and announcements → Event, Announcement
 - Enable join club action → Club Membership Controller
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Class: Club Dashboard**Description:**

Represents the management interface for club leaders.

Responsibilities / Collaborators:

- Display club management tools → Event Management Controller
 - Manage members → Club Membership Controller
 - Manage announcements → Announcement Controller
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Class: Announcement List View

Description:

Represents the interface for viewing club announcements.

Responsibilities / Collaborators:

- Display announcements → Announcement
 - Refresh announcement list → Announcement Controller
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Class: Admin Panel

Description:

Represents the administrative interface for system oversight.

Responsibilities / Collaborators:

- Display moderation and approval tools → Admin
- Support administrative actions → Club Approval Controller, Content Moderation Controller

Control Objects

Class: Authentication Controller

Description:

Controls authentication and access to the system.

Responsibilities / Collaborators:

- Validate user credentials → User
 - Manage login and logout → Login Page
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Class: Event Browsing Controller

Description:

Controls retrieval and filtering of events.

Responsibilities / Collaborators:

- Retrieve event listings → Event
 - Apply search and filter criteria → Event Filter Panel
 - Provide results for display → Event List View
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Class: Event Registration Controller

Description:

Controls student registration for events.

Responsibilities / Collaborators:

- Register students for events → Student, Event
 - Enforce capacity rules → Event Registration
 - Update event details view → Event Details View
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Class: Event Management Controller

Description:

Controls creation and management of events.

Responsibilities / Collaborators:

- Create, edit, and cancel events → Event
- Associate events with clubs → Club

- Support club leader actions → Club Dashboard
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Class: Club Browsing Controller

Description:

Controls retrieval and display of club information.

Responsibilities / Collaborators:

- Retrieve club listings → Club
 - Provide club information → Club Browse View, Club Page View
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Class: Club Membership Controller

Description:

Controls club membership processing.

Responsibilities / Collaborators:

- Process membership requests → Membership Request
 - Approve or reject members → Club Membership
 - Support club leader actions → Club Dashboard
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Class: Announcement Controller

Description:

Controls creation and retrieval of announcements.

Responsibilities / Collaborators:

- Create and publish announcements → Announcement
 - Retrieve announcements for display → Announcement List View
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Class: Club Approval Controller

Description:

Controls approval of new club registrations.

Responsibilities / Collaborators:

- Review club submissions → Club
 - Approve or reject clubs → Admin
 - Support administrative actions → Admin Panel
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Class: Content Moderation Controller

Description:

Controls moderation of reported content.

Responsibilities / Collaborators:

- Review flagged content → Content
- Remove or resolve violations → Admin
- Support administrative actions → Admin Panel

